In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Advanced)-2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – V
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A, C**	**A, C, D**	**A, B, C**
2.	**A, C, D**	**B, D**	**A, C**
3.	**B, D**	**B, C**	**A, B, C**
4.	**A, C**	**A, C**	**A, B, C**
5.	**A, C, D**	**A, B, C, D**	**A, B, C, D**
6.	**A, B**	**A, D**	**A, C**
7.	**C**	**B, C**	**A, B, C**
8.	**A**	**B, D**	**A, B**
9.	**D**	**A, C**	**C, D**
10.	**A, B, D**	**A, B, C, D**	**A, B, C**
1.	**6**	**7**	**2**
2.	**2**	**6**	**7**
3.	**3**	**4**	**6**
4.	**5**	**8**	**4**
5.	**4**	**3**	**2**
6.	**8**	**8**	**2**
7.	**1**	**2**	**2**
8.	**4**	**8**	**8**
9.	**2**	**3**	**4**
10.	**2**	**3**	**2**

Physics

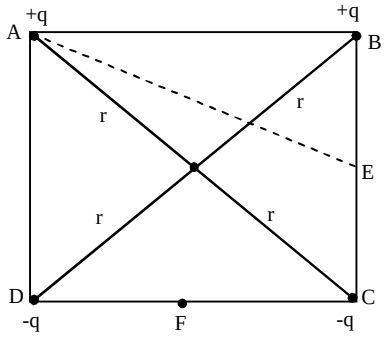
PART – I

SECTION – A

1. **A, C**

Refer to figure. potential at O is

$$V_0 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r}+\frac{q}{r}-\frac{q}{r}-\frac{q}{r}\right)=0$$



Refer to fig. again $(AE)^2 = a^2 + \left(\frac{a}{2}\right)^2$

$= \frac{5a^2}{4}$, giving $AE - \frac{\sqrt{5}a}{2}$. Similarly $DE = \frac{\sqrt{5}a}{2}$ and $BE = \frac{a}{2}, CE = \frac{a}{2}$. Potential at E is

$$V_E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{AE}+\frac{q}{BE}-\frac{q}{DE}-\frac{q}{CE}\right)=0$$

Work done in carrying a charge – e from O to E is

$$W = -e(V_E - V_0)$$
$$= -e(0-0) = 0$$

Potential at F is

$$V_F = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{AF}+\frac{q}{BF}-\frac{q}{DF}-\frac{q}{CF}\right)$$

Now, $AF = BF = \sqrt{5}\ a/2$ and CF = DF = a/2

Putting these values, we get

$$V_F = \frac{q}{\pi\varepsilon_0 a}\left(\frac{1}{\sqrt{5}}-1\right)$$

Work done is carrying a charge – e from O to F is

$$W = -e\left(V_F - V_0\right) = -eV_F$$
$$= \frac{qe}{\pi\varepsilon_0 a}\left(1-\frac{1}{\sqrt{5}}\right)$$

Hence the correct choice is (a) and (c).

2. **A, C, D**

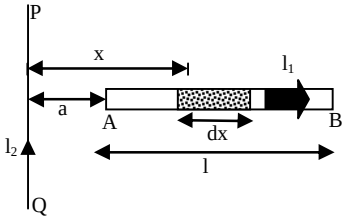
In the steady state, no current flows through the branch ABC containing the capacitors. Thus all the current flows through ADC. So choice (a) is correct. Current through R_1 = current through R_2 = I = 12/24 = 0.5 A. Now p.d. across A and C = 12 V. Therefore $Q = V_1C_1 = V_2C_2$. Thus

$\frac{V_1}{V_2} = \frac{C_2}{C_1}$ =2. Also $V_1 + V_2 = 12$. Therefore, V_1 = 8 V and V_2 = 4V. Thus

$Q = C_1V_1 = 10\mu F \times 8V = 80\mu C =$ charge on C_2. Thus the correct choices are (a), (c) and (d)

3. **B, D**

Consider a small element of length dx of the rod at a distance x from wire PQ as shown in figure. The magnetic field use to wire PQ at the element is



$$dF = BI_1\, dx = \frac{\mu_0 I_1 I_2 dx}{2\pi x}$$

$\therefore$ Total force experienced by the rod is

$$F = \int dF$$

$$= \frac{\mu_0 I_1 I_2}{2\pi_2} \int_{x=a}^{x=(a+l)} \frac{dx}{x}$$

$$= \frac{\mu_0 I_1 I_2}{2\pi} \log_e \left(\frac{a+l}{a} \right)$$

Now, the magnetic field B is not uniform; it decreases as we go from and A to end B of the rod. Hence the rod will also experience a torque. Thus the correct choices are (b) and (d).

4. **A, C**

$\vec{v} = v\hat{e}_r + r\omega\hat{e}_\theta,\ t = \frac{\theta}{\omega},\ \vec{r} = \frac{v\theta}{\omega}\hat{e}$

5. **A, C, D**

Using Gauss theorem.

6. **A, B**

(a,b) as $E \propto \frac{1}{n^2}, P \propto \frac{1}{n}$ & $W \propto \frac{1}{n^3}$

7. **C**

$^{17}N_7 \longrightarrow {}^{6}Li_3 + 2\ {}_2He^4 + {}_0n^1 + 2\gamma$

8. **A**

From black body radiation concept.

9. **D**

The resistances of bulbs B_1, B_2 and B_3 respectively are

$$R_1 = \frac{V^2}{W_1} = \frac{(250)^2}{100} = 625\Omega$$

$$R_2 = \frac{(250)^2}{60} = 1042\Omega = R_3$$

Voltage across B_3 is $V_3 = 250$ V

Voltage across B_1 is $V_1 = \frac{VR_1}{(R_1 + R_2)}$

$$= \frac{250 \times 625}{(625 + 1042)} = 93.7V$$

Voltage across B_2 is $V_2 = 250 - 93.7 = 156.3V$

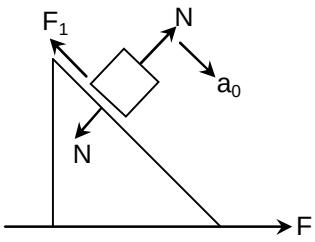
$$W_1 = \frac{V_1^2}{R_1} = \frac{(93.7)^2}{625} = 14W$$

Power output $W_2 = \frac{(156.3)^2}{1042} = 23W$

$$W_3 = \frac{(250)^2}{1042} = 60W$$

Hence $W_1 < W_2 < W_3$, which is choice (D)

10. **A, B, D**



$a_0 = g\sin\theta$

$F = N\sin\theta,\ N = mg\cos\theta$

$\Rightarrow F = \frac{1}{2}mg\sin 2\theta$

SECTION – C

1. **6**

$A_1V_1 = A_2V_2$

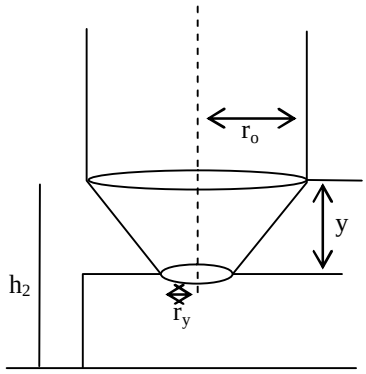
$V_y = \left(\frac{r_0}{r_y}\right)^2 u_0$

By Bernoulli's theorem

$\frac{p}{\rho} + \frac{1}{2}V_y^2 + gh_1 = \frac{p}{\rho} + \frac{1}{2}u_0^2 + gh_2$

$\Rightarrow r_y = r_0\left(1 + \frac{2gy}{u_0^2}\right)^{-1/4} = r_0[1 + 2y]^{-1/4}$

$\Rightarrow \eta + \alpha = 6$



2. **2**

$\bar{\tau}_A = 0$

$m_2 g\ell_2 \sin 45^\circ = m_1 g\ell_1 \sin 45^\circ$ ..... (1)

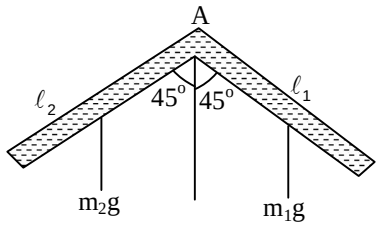
$m_1 = \rho_1\ell_1,\ m_2 = \rho_2\ell_2$

From (1),

$\rho_1\ell_1^2 = \rho_2\ell_2^2$

$\eta = \frac{-\rho_1}{\rho_2} = \left(\frac{5}{3}\right)^2 = \frac{25}{9}$

$x = \frac{18}{25}\eta = 2$



3. **3**

In steady state heat flow will be same across each sheet.

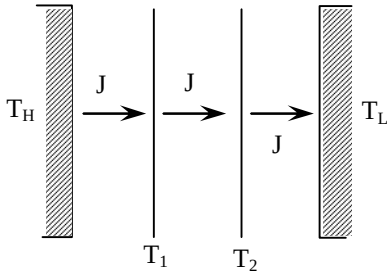
Now $J = \sigma\left(T_H^4 - T_1^4\right)A$

$J = \sigma\left(T_1^4 - T_2^4\right)A$

$J = \sigma\left(T_2^4 - T_L^4\right)A$

Now $3J = \sigma\left(T_H^4 - T_L^4\right) = J_o$

$\frac{J_o}{J} = 3$



4. **5**

Speed of ions

$V = \frac{E_1}{B_1} = 5000\ m\ s^{-1}$

$S_1S_2 = 2(r_2 - r_1) = 2\left[\frac{m_2 v}{qB} - \frac{m_1 v}{qB}\right]$

$= \frac{2\times 5000}{B}\times(22-20)\frac{amu}{e}$

$= \frac{10^4}{8\times 10^{-2}}\times 2\times 10^{-8}$

$d = \frac{1}{4}\times 10^{-2} m = 2.5mm$

5. **4**

For plane surface.

$\frac{n_1}{-mR} + \frac{n_2}{v} = \frac{n_2 - n_1}{R_1}$

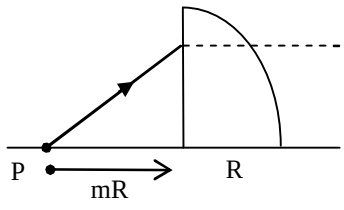
$n_1 = 1, n_2 = 1.5$ 1 $R_1 = \infty \Rightarrow v = -1.5mR$

For curve surface

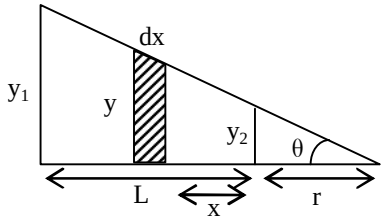
$\frac{n_1}{-(1.5mR + R)} + \frac{n_2}{\infty} = \frac{n_2 - n_1}{R}$

$n_2 = 1, n_1 = 1.5$

$\Rightarrow m = \frac{4}{3}, 3m = 4$



6. **8**



$$dR = \rho\frac{dx}{wy} \Rightarrow \frac{y}{x+r} = \frac{y_1}{L+r}$$

$$R = \int_0^L \rho\frac{dx}{wy} = \frac{\rho(L+r)}{wy_1}\ln\frac{L+r}{r}$$

Now $r = \frac{y_2}{y_1 - y_2}L$ $R = \frac{\rho L}{w(y_1 - y_2)}\ln\frac{y_1}{y_2} = \frac{36\rho}{L}\ln 2$

7. **1**

$$x_{cm} = \frac{A_1x_1 - A_2x_2}{A_1 - A_2}$$

$$A_1 = \frac{1}{2}\times 6\times 9 = 27cm^2$$

$$A_2 = \frac{1}{2}\times 6\times 3 = 9cm^2$$

$$X_1 = 0, X_2 = -2$$

$$X_{cm} = \frac{27\times 0 - 9\times -2}{27-9} = 1cm$$

8. **4**

$$F = v\frac{dm}{dt} = \lambda v^2$$

4N

9. **2**

$$V_{AB} = X$$

$$R_1 + \frac{R_2x}{R_2 + x} = x$$

$$V_{AB} = 2V_{cd}$$

$$\therefore V_{AC} = V_{CD}$$

$$\therefore R_1 = \frac{R_2x}{R_2 + x}$$

Now $2R_x = x$

$$R_1 = \frac{R_1R_2}{R_2 + 2R_1} \Rightarrow \frac{R_2}{R_1} = 2$$

10. **2**

$$E_c = \frac{\delta}{4\varepsilon_o} = 2N/C$$

Chemistry

PART – II

SECTION – A

1. **A, C, D**
ΔH is a state function so it is independent of path.

2. **B, D**
The yield of product will depend on stability of the carbocation. Carbocation corresponding to (y) is more stable than (z) as well as (x).

3. **B, C**
$KNO_3 \longrightarrow KNO_2 + O_2$
$NaNO_3 \longrightarrow NaNO_2 + O_2$
$Cu(NO_3)_2 \longrightarrow CuO + NO_2$
$Pb(NO_3)_2 \longrightarrow PbO + NO_2$

4. **A, C**
Fact.

5. **A, B, C, D**
(A) bond cleavage takes place as a result of reductive ozonolysis.
(B) epoxidation
(C) syn addition
(D) oxidative cleavages in basic medium

6. **A, D**
Fact.

7. **B, C**
Highly electropositive metal are extracted by electrolytic reduction.

8. **B, D**
Fact.

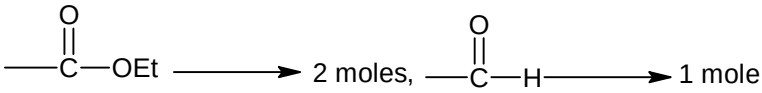
9. **A, C**
Rotation is restricted about $\pi -$ bond .

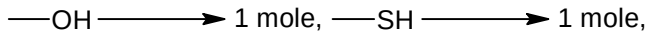
10. **A, B, C, D**
Fact.

SECTION – C

1. **7**
I_3^+, XeF_4, C_2F_4
$SO_3, NOCl, ClF_3$
$F_2CO,$

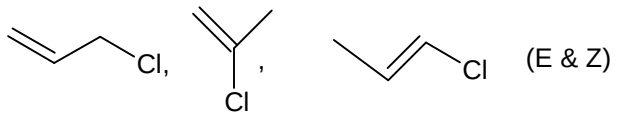
2. **6**





Epoxide – 1 mole so total 6 moles.

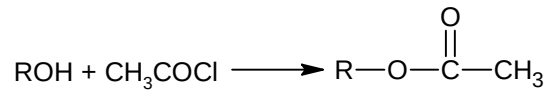
3. **4**



4. **8**

$\left[Pt(gly)_2Cl_2\right]$ and $\left[Co(en)_2Cl_2\right]$ have 6 & 2 optically active isomers respectively.

5. **3**



In the above reaction molecule wt. of product is increased by 42 units.
So if n, - OH groups were present in the compound then increase in molecule wt. will be 42 n
increase in mass of compound $= 218 - 92 = 126$
$42n = 126 \Rightarrow n = 3$

6. **8**

$\Delta H_R^0 = 2\times\Delta H_f^0(NH_3) - 3\times\Delta H_f^0(H_2) - \Delta H_f^0(H_2) = -20\text{Kcal}$

$\Delta S_R^0 = 2\times\Delta S_f^0(NH_3) - 3\times\Delta S_f^0(H_2) - \Delta S_f^0(N_2)$

$= 2\times45 - 3\times30 - 40 = -40 \text{ cal} = -0.04 \text{ Kcal}$

$\Delta G^0 = \Delta H^0 - T\Delta S^0 = -8 \text{ K cal}$

7. **2**

$2NO + O_2 \longrightarrow 2NO_2$

2 0.5 0

$2-1$ 0 1

$n_i = 2.5,\ n_f = 2$

$\Delta n = 0.5$

$\Delta P = \dfrac{\Delta nRT}{v} = \dfrac{0.5\times0.0821\times300}{6.25} = 2$

8. **8**

$\Delta E = \dfrac{3}{4}\times0.85\text{eV}$

$0.85\left(1-\dfrac{1}{4}\right) = 13.6\left(\dfrac{1}{4^2} - \dfrac{1}{n^2}\right)$

$n = 8$

9. **3**

$S - e^- = 8$

$P - e = 18$

$Pe^- - Se^- = 10$

$de^- = 10$

$At.No = 36$

$6 - 3 = 3$

10. **3**

$$3A(g) \rightarrow 2B(g) + 2C(g)$$

t = 0	3P	0	0
t = 9	$3P-3P_1$	$2P_1$	$2P_1$
$t = \infty$	0	2P	2P

Total pressure after long time = 4P = 400 $\Rightarrow$ P = 100

Pressure after 9 minutes = $3P + P_1 = 387.5 \Rightarrow P_1 = 87.5$

Pressure of A becomes one-eighth in 9 minutes. That means $3t_{1/2} = 9 \Rightarrow t_{1/2} = 3$

Mathematics

PART – III

SECTION – A

1. **A, B, C**

Replace x by 2, we get

$2f(2)+2f\left(\frac{1}{2}\right)-2f(1)=4 \quad \ldots\ldots\ldots(i)$

$\Rightarrow f(2)+f\left(\frac{1}{2}\right)=2+f(1)$

Replace x by 1, $f(1)=-1 \quad \ldots.(ii)$

Replace x by $\frac{1}{2}, 2f\left(\frac{1}{2}\right)+\frac{1}{2}f(2)+2=\frac{5}{2} \quad \ldots\ldots(iii)$

Solving (i) & (iii) $f\left(\frac{1}{2}\right)=0, f(2)=1$

2. **A, C**

$\sin B=\frac{1}{\sqrt{2}},\ \tan\left(\frac{B}{2}\right)=\left(\sqrt{2}-1\right)$

$\&\ \sin C=\sin\left(135^{o}-A\right)=\frac{\sqrt{2}(\sin A+\cos A)}{2}$

$r=\frac{c+a-b}{2}\tan\left(\frac{\mathbf{B}}{2}\right)$

$=R\left(\sqrt{2}-1\right)(\sin A+\sin C-\sin B)$

$OI=R^{2}-2Rr$

$OI^{2}=R^{2}\left(1-2\sin A-2\sin C+2\sin B\left(\sqrt{2}-1\right)\right) \quad \ldots\ldots\ldots\ldots(ii)$

$\&\ \sqrt{2}OI=AB-AC$

$OI^{2}=\left(C-2(\sin A+\sin C-\sin B)\right)\left(\sqrt{2}-1\right)\ldots\ldots\ldots(iii)$

From (i) & (ii), we obtain

$2(\sin C-\sin B)^{2}=\left(1-2(\sin A+\sin C-\sin B)\left(\sqrt{2}-1\right)\right)$

$\Rightarrow 1-2(\sin C-\sin B)^{2}=(\sin A+\sin C-\sin B)\left(\sqrt{2}-1\right)$

$\Rightarrow 1-2\sin^{2}C+2\sqrt{2}\sin C-1=2(\sin A+\sin C)\left(\sqrt{2}-1\right)-\left(2-\sqrt{2}\right)$

$\Rightarrow -(\sin A+\cos A)+2(\sin A+\cos A)=\left(2\sqrt{2}-2\right)\sin A+\left(2-\sqrt{2}\right)(\sin A+\cos A)$

$<-\left(2-\sqrt{2}\right)$

$\Rightarrow -1-2\sin A\cos A=\left(\sqrt{2}-2\right)\sin A-\sqrt{2}\cos B-2\left(2-\sqrt{2}\right)$

$\Rightarrow \left(\sqrt{2}\sin A-1\right)\left(\sqrt{2}\cos A-\sqrt{2}+1\right)=0$

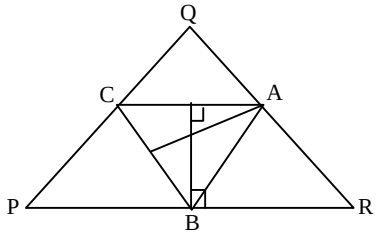
$\Rightarrow \sin A=\frac{1}{\sqrt{2}}\ \&\ \cos A=\frac{1-\sqrt{2}}{\sqrt{2}} \quad \Rightarrow \sin A=\frac{\sqrt{4\sqrt{2}-2}}{2}$

3. **A, B, C**

$AC \parallel PR$ & $2AC = PR$

So, ABPC is a parallelogram comparing the coordinates of midpoints of diagonal, we get

$P(-a,b,c)\,Q(a,-b,c)$ & $R(a,b,-c)$.



Also AD & AP are medians of ΔABC and ΔPQR respectively.

So, centroids are coinciding. The perpendicular bisector of PR is also perpendicular to AC.

$\therefore$ the circumcenter of ΔPQR is orthocentre of ΔABC

$\text{area}(\Delta PQR) = 4\,\text{area}(\Delta ABC)$

$= 4\sqrt{(OAB)^2 + (OBC)^2} + (OAC)^2$

Where OAB is the area of projection of ΔABC on the plane XOZ etc.

4. **A, B, C**

$(\bar{x}\times\bar{y})\times\bar{b} = \bar{a}\times\bar{b}$

$\Rightarrow (\bar{x}.\,\bar{b}).\bar{y} - (\bar{y}\,.\,\bar{b})\bar{x} = \bar{a}\times\bar{b}$

$\Rightarrow \gamma.\bar{y} - (\bar{y}\,.\,\bar{b}).\bar{x} = \bar{a}\times\bar{b} \qquad (\because \bar{x}.\bar{b} = \gamma)$

$\bar{y}.\bar{b} = \bar{y}.(\bar{y}\times\bar{z}) = 0$

$\Rightarrow \bar{y} = \dfrac{\bar{a}\times\bar{b}}{\gamma}$

$\Rightarrow (\bar{x}\times\bar{y})\times\bar{y} = \bar{a}\times\bar{y}$

$\Rightarrow (\bar{x}.\bar{y}).\bar{y} - (\bar{y}.\bar{y})\bar{x} = \bar{a}\times\bar{y}$

$\Rightarrow \bar{y} - |\bar{y}|^2\,\bar{x} = \bar{a}\times\bar{y} \qquad (\because \bar{x}.\bar{y} = 1)$

$\Rightarrow \bar{x} = \dfrac{1}{y^2}[\bar{y} - \bar{a}\times\bar{y}]$

$(\bar{y}\times\bar{z})\times\bar{y} = \bar{b}\times\bar{y}$

$\Rightarrow y^2\bar{z} - (\bar{z}.\,\bar{y}).\bar{y} = \bar{b}\times\bar{y} \;\Rightarrow y^2\bar{z} - \bar{y} = \bar{b}\times\bar{y}$

$\Rightarrow \bar{z} = \dfrac{1}{y^2}[\bar{y} + \bar{b}\times\bar{y}]$

5. **A, B, C, D**

$I = \int_0^1 {}^{207}C_7\,\dfrac{x^{200}}{II}\,\dfrac{(1-x)^7\,dx}{I}$

$= {}^{207}C_7\left[(1-x)^7\,,\dfrac{x^{201}}{201}\right]_0^1 + \dfrac{7}{207}\int_0^1 (1-x)^6\,.x^{201}dx$

$= {}^{207}C_7\,\dfrac{7}{201}\int_0^1 (1-x)^6\,x^{201}dx$

Similarly by applying integration by part 6 times we get

$$I = {}^{207}C_7 \cdot \frac{7.6.5.4.3.2.1}{201.202.203.....207}\int_0^1 x^{207}dx$$

$$= \frac{207!}{7!200!}.\frac{7!}{201!202.203}.....\frac{1}{207....208}$$

$$= \frac{1}{208} = \frac{1}{k}$$

6. **A, C**

$$f(x) = kx$$

$$1.\int_0^1 kxdx + 2\int_0^2 kxdx$$

$$+3\int_0^3 kxdx + 4\int_0^4 kxdx + 5.\int_0^5 kxdx$$

$$= \frac{k}{2}\left[1^3 + 2^3 + 3^3 + 4^3 + 5^3\right]$$

$$= \frac{k}{2}\left[\frac{5\times 6}{3}\right]^2$$

$$= \frac{225K}{2}$$

$$\text{given} \frac{225K}{2} = 450$$

$$\Rightarrow k = 4$$

7. **A, B, C**

$$\bar{a}.\left(\bar{b} - a\right) - \bar{c}\left(\bar{b} - \bar{a}\right) = 0$$

$$\overline{AC}.\ \overline{AB} = 0$$

Also $a + c = 2b - (i)$ & $b^2 + c^2 = a^2$

$$b^2 = (a-c)(a+c) = 2b(a-c)$$

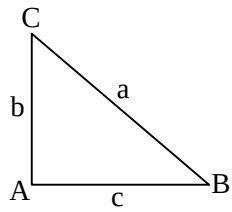
$$\Rightarrow b = 2(a-c) - (ii)$$

From (i) & (ii) we get

$$\frac{a}{5} = \frac{b}{4} = \frac{c}{3} = \lambda$$

$$\because \text{Inradius } = 1 \Rightarrow \frac{6\lambda^2}{6\lambda} = 1 \Rightarrow \lambda = 1$$

$$\therefore a = 5, b = 4, c = 3$$



8. **A, B**

At the point of contact of two functions their value are equal and their derivatives are equal.

$f(x) = g(x)$ & $f'(x) = g'(x)$

$ax^3 + 4x = \frac{1}{x}$ & $3ax^2 + 4 = -\frac{1}{x^2}$

$\Rightarrow x^2 = \frac{1}{2}$

$\Rightarrow a = -4$

$\therefore P\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$ & $Q\left(\frac{-1}{\sqrt{2}}, -\sqrt{2}\right)$

Area of triangle formed by any tangent to rectangular hyperbola $xy = c^2$ with coordinate axis equal to $2c^2$

$\therefore$ Area $= 2$

9. **C, D**

$$\int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}dx$$

$$= \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cot^{-1}\left(\frac{-x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx$$

$$2I = \int_{-1}^{1} \cot^{-1}\frac{1}{\sqrt{1-x^2}}\left(\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}} + \pi - \cot^{-1}\frac{x}{\sqrt{1-(x)^{2|x|}}}\right)dx$$

$$2I = \pi\int_{-1}^{1} \cot^{-1}\frac{1}{\sqrt{1-x^2}}dx$$

$$2I = \pi\int_{-1}^{1} \tan^{-1}\sqrt{1-x^2}\,dx$$

Integrating by parts

$$I = \frac{\pi}{2}\left[x\tan^{-1}\sqrt{1-x^2}\right]_0^1 - \pi\int_0^1 \frac{x}{1+1-x^2}\left(\frac{-x}{\sqrt{1-x^2}}\right)dx$$

$$I = \frac{\pi}{2}\int_0^1 \frac{x^2dx}{\left(2-x^2\right)\sqrt{1-x^2}}$$ put $x = \sin\theta$, $dx = \cos\theta\, d\theta$

$$= \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin^2\theta}{2-\sin^2\theta}d\theta$$

$$= \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin^2\theta - 2 + 2}{2-\sin^2\theta}d\theta$$

$$= \frac{-\pi}{2}\int_0^{\pi/2} d\theta + \pi\int_0^{\pi/2} \frac{d\theta}{2-\sin^2\theta}$$

$$= \frac{-\pi^2}{4} + \pi\int_0^{\pi/2} \frac{\sec^2\theta d\theta}{2\tan^2\theta + 2 - \tan^2\theta}$$

$$= \frac{-\pi^2}{4} + \pi\int_0^\infty \frac{dk}{k^2+2} \qquad (\tan\theta = k)$$

$$= \frac{-\pi^2}{4} + \pi.\frac{1}{\sqrt{2}}\left[\tan^{-1}\frac{k}{\sqrt{2}}\right]_0^\infty$$

$$= \frac{-\pi^2}{4} + \frac{\pi^2}{2\sqrt{2}}$$

$$= \pi^2\left(\frac{-1+\sqrt{2}}{4}\right)$$

10. **A, B, C**

$$A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^3 = 0$$

$$(A+I)^3 = A^3 + 3A^2 + 3A + I$$

$$(A+I)^{50} = 50A + I$$

$$(A+I)^{50} - 50A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

SECTION – C

1. **2**

$$\frac{\tan\theta}{\cos 2\theta} = \frac{\tan\theta(1+\tan^2\theta)}{1-\tan^2\theta} = \tan 2\theta - \tan\theta$$

$$\therefore (\tan 2 - \tan 1) + (\tan 4 - \tan 2) + ... + \tan(2048 - \tan 1024)$$

$$= \tan 2048 - \tan 1$$

2. **7**

$\sin A = \sin 2B \Rightarrow \sin A = 2\sin B\cos B$

Also, sinC = sin3B = 3 sinB – 4 $\sin^3$B

$\therefore a = 2b\cos B, c = b(3 - 4\sin^2 B) = b(4\cos^2 B - 1)$

hence $a^2 = b(b+c)$

A triangle of smallest perimeter means $\gcd(a,b,c) = 1$.

In fact $\gcd(b,c) = 1$ since any common factor of b, c would be a factor of a as well.

$\because$ A perfect square a^2 is being expressed as the product of two relatively prime integers b and c, it must be the case that both b & b + c are perfect squares. Thus for some integer m & n with $\gcd(m, n) = 1$ we have

$b = m^2$ & $b + c = n^2, a = mn$

$2\cos B = \dfrac{n}{m} = \dfrac{a}{b}$

$\because \angle C > 90^o \Rightarrow O < B < 30^o$

$\&\sqrt{3} < 2\cos B = \dfrac{n}{m} < 2$

The smallest value of (m, n) that satisfies the above mentioned conditions are 4 and 7.

3. **6**

$$\sum_{k=0}^{7}\left(\frac{{}^{7}C_k}{\lfloor 14}\lfloor k\ \lfloor 14-k\sum_{r=K}^{14}\frac{\lfloor r}{\lfloor k\ \lfloor r-k}\frac{\lfloor 14}{\lfloor r\ \lfloor 14-r}\right)$$

$$=\sum_{k=0}^{7}\left({}^{7}C_k\sum_{r=k}^{14}{}^{14-K}C_{7-k}\right)=\sum_{k=0}^{7}{}^{7}C_k\,2^{14-k}$$

$$=2^{14}\sum_{k=0}^{7}{}^{7}C_k\left(\frac{1}{2}\right)^k=6^7>7^6$$

4. **4**

$$I=\int_0^1\frac{\sin^{-1}\sqrt{x}}{x^2-x+1}dx \quad .....(i)$$

$$=\int_0^1\frac{\sin^{-1}\sqrt{1-x}}{(1-x)^2-(1-x)+1}dx$$

$$=\int_0^1\frac{\cos^{-1}\sqrt{x}}{x^2-x+1}dx \qquad .....(ii)$$

Adding (i) & (ii)

$$2I=\frac{\pi}{2}\int_0^1\frac{1}{x^2-x+1}dx$$

$$2I=\frac{\pi}{2}.\frac{1}{\left(\frac{\sqrt{3}}{2}\right)}\left[\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)\right]_0^1$$

$$2I=\frac{\pi^2}{3\sqrt{3}}$$

$$I=\frac{\pi^2}{\sqrt{108}}$$

$n = 108$

5. **2**

$z = \cos\theta(\cos\theta + i\sin\theta)$

$$\sum_{k=1}^{n}z^k=\frac{z-z^{n+1}}{1-z}$$

$$\sum_{k=1}^{n}\cos^k\theta(\cos k\theta+i\sin k\theta)$$

$$=\frac{\cos\theta(\cos\theta+i\sin\theta)-\cos^{n+1}\theta(\cos(n+1)\theta+i\sin(n+1)\theta)}{\sin^2\theta-i\cos\theta\sin\theta}$$

$$= \frac{i\cos\theta(\cos\theta + i\sin\theta) - \cos^{n+1}\theta(\cos(n+1)\theta + i\sin(n+1)\theta)}{\sin\theta(\cos\theta + i\sin\theta)}$$

$$= i\left(\cot\theta - \frac{\cos^{n+1}\theta(\cos n\theta + i\sin n\theta)}{\sin\theta}\right)$$

$$= \frac{\sin n\theta\cos^{n+1}\theta}{\sin\theta} + i\left(\cot\theta - \frac{\cos^{n+1}\theta\cos n\theta}{\sin\theta}\right)$$

$$\sum_{k=1}^{n}(\cos\theta)^k\cos k\theta = \frac{\sin n\theta\cos^{n+1}\theta}{\sin\theta}$$

For $\theta = 30^\circ$

$$= 2\sin\frac{n\pi}{6}\left(\frac{\sqrt{3}}{2}\right)^{n+1}$$

6. **2**

In one stage the tourist traverses the complex number

$$x = 100 + 100\bar{z} + 100\bar{z}^{-2} = 100 - 100\sqrt{3}i$$

where $z = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$

In 2017 stages, the tourist traverses the complex number

$$W = x + xz + xz^2 + ... + xz^{2016}$$

$$= x\left(\frac{1 - z^{2017}}{1 - z}\right) =$$

$$= xz$$

$$|W| = |xz| = 200$$

7. **2**

$$a_{ij} = 0 \forall i \neq j$$

$$a_{ij} = (n-1)^2 + i \forall i = j$$

$\therefore$ sum of all the elements of A_n

$$= \sum_{i=1}^{2n-1}\left[(n-1)^2 + i\right]$$

$$= (2n-1)(n-1)^2 + (2n-1)n$$

$$= 2n^3 - 3n^2 + 3n + 1$$

$$= n^3 + (n-1)^3$$

$$T_n = (-1)^n\left[n^3 + (n-1)^3\right]$$

$$= (-1)^n n^3 - (-1)^{n-1}(n-1)^3$$

$$= V_n - V_{n-1}$$

$$\Rightarrow \sum_{n=1}^{102} T_n = \sum_{n=1}^{102}(V_n - V_{n-1}) = V_{102} - V_0 = (102)^3$$

$$\Rightarrow \left[\frac{\sum_{n=1}^{-102} T_n}{520200}\right] = 2$$

8. **8**

$$f'(x)=\lim_{x\to\infty}\frac{f(x+h)-f(x+0)}{h}$$

$$=\lim_{x\to\infty}\frac{2f(x)+xf(h)+h\sqrt{f(x)}-2f(x)-xf(0)+0\sqrt{f(x)}}{h}$$

As f (0) = 0

$$\Rightarrow \lim_{h\to0} x\left(\frac{f(h)-f(0)}{h-0}\right)+\sqrt{f(x)}$$

$$f'(x)=\sqrt{f(x)}$$

$$\int\frac{f'(x)}{\sqrt{f(x)}}dx=\int dx \quad \Rightarrow 2\sqrt{f(x)}=x+C$$

$$f(x)=\frac{x^2}{4}$$

when $\alpha=0$, area is minimum

Required min area = $2\int_0^9 2\sqrt{y}dy$

$= 72$ sq. unit

9. **4**

$$f'\left(\frac{5}{2}-x\right)=f'\left(\frac{5}{2}+x\right)$$

Replacing x by $\frac{5}{2}-x$

$$f'(x)=f'(5-x)\Rightarrow F(x)=-F(5-x)+C$$

$$\Rightarrow C=f(x)+f(5-x)$$

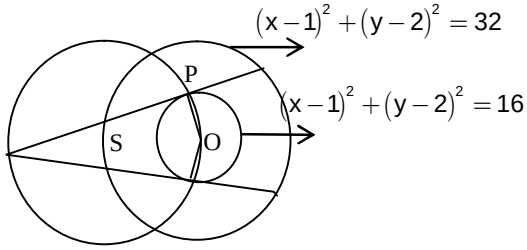
$$\Rightarrow C=8$$

$$I=\int_2^3 f(x)dx=\int_2^3 f(5-x)dx$$

$$\Rightarrow 2I=\int_2^3 f(x)+f(5-x)dx$$

$$=\int_2^3 8dx \quad \Rightarrow I=4$$

10. **2**



$TS = OT - SO = 12 - 4\sqrt{2}$